

A Note on Contractive Update Rules

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1 Setting

Throughout this note we work in a finite dimensional real vector space with the usual inner product. Let $V = \mathbb{R}^n$ with norm $\|x\| = \sqrt{\langle x, x \rangle}$, and let $f: V \rightarrow V$ be a map with $f(0) = 0$. For $x_0 \in V$ we define the sequence $x_{k+1} = f(x_k)$ for $k \geq 0$, and we ask when $x_k \rightarrow 0$ as $k \rightarrow \infty$ for every choice of x_0 with $\|x_0\| \leq r$ and some fixed $r > 0$.

Definition 1. A map f is called contractive on a set $S \subseteq V$ if there is a constant $0 \leq q < 1$ such that $\|f(x) - f(y)\| \leq q \|x - y\|$ for all $x, y \in S$.

The simplest contractive maps are linear. If $f(x) = Ax$ for a matrix A then

$$\|x_k\| = \|A^k x_0\| \leq \|A\|^k \|x_0\|, \quad (1)$$

so the iteration converges whenever the operator norm of the matrix is smaller than one. A concrete example in two dimensions is the matrix

$$A = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{3} \end{pmatrix}, \quad A^k = \begin{pmatrix} 2^{-k} & c_k \\ 0 & 3^{-k} \end{pmatrix}, \quad (2)$$

where $c_k = \frac{3}{2} (2^{-k} - 3^{-k})$ for all $k \geq 0$.

Theorem 2. *Let f be contractive on the closed ball $B_r = \{x : \|x\| \leq r\}$ with constant q , and suppose $f(B_r) \subseteq B_r$. Then for every $x_0 \in B_r$ the sequence of iterates satisfies $\|x_k\| \leq q^k r$, and in particular it converges to the origin.*

2 Proofs

The proof of the main theorem rests on a simple estimate that we record separately, because it is used again in the next section.

Lemma 3. *Under the assumptions of Theorem 2, for all $k \geq 0$ we have*

$$\|x_{k+1}\| = \|f(x_k) - f(0)\| \tag{3}$$

$$\leq q \|x_k - 0\| \tag{4}$$

$$= q \|x_k\|. \tag{5}$$

Proof. The first line uses $f(0) = 0$, the second is the contraction property on B_r , which applies because both x_k and 0 lie in the ball, and the third is immediate. $\square$

Proof of Theorem 2. By induction on k and Lemma 3, $\|x_k\| \leq q^k \|x_0\| \leq q^k r$. Since $0 \leq q < 1$ we have $q^k \rightarrow 0$, hence $x_k \rightarrow 0$. $\square$

For the example in (2), the operator norm satisfies $\|A\|_2 \approx 0.557$, which gives the explicit rate

$$\|x_k\| \leq (0.557)^k \|x_0\| \quad \text{and} \quad \sum_{k=0}^{\infty} \|x_k\| \leq \frac{\|x_0\|}{1 - 0.557}. \tag{6}$$

3 A piecewise rule

Consider now the piecewise definition

$$g(x) = \begin{cases} \frac{1}{2}x & \text{if } \|x\| \leq 1, \\ \frac{x}{2\|x\|^2} & \text{otherwise.} \end{cases} \tag{7}$$

Inside the unit ball this map halves every vector, and outside it the image lies strictly inside the ball, so every orbit enters the ball after one step and then converges at the rate given by (1) with $q = \frac{1}{2}$.